

Important dates

Date	Note
09/28	Due: A1 – project brief and pilot data
09/29	Consultation 1 – collection gate
10/20	Midterm review
10/22	Midterm exam
11/04	Due: A2 – analysis-ready and EDA
11/05	Consultation 2 – dataset gate
12/08	Consultation 3 – presentation gate
12/10, 12/15	Final presentation
12/17	Due: A3 – final report

Schedule: first half of semester

Date	Topics
w1 09/01	Course introduction & house rules
09/03	Group project introduction
w2 09/08	Web scraping I
09/10	Web scraping II
w3 09/15	Web scraping III
09/17	Mock interview 1
09/22	Web scraping IV
w4 09/24	Working with generative AI
w5 09/29	Consultation 1 – collection gate
10/01	National Day Holiday (no class)
w6 10/06	
10/08	Unstructured data analysis I
w7 10/13	Unstructured data analysis II
10/15	Unstructured data analysis III
w8 10/20	Midterm review
10/22	Midterm exam



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MGS 4701 W01 | Fall 2026

Integrative Application of Business Analytics

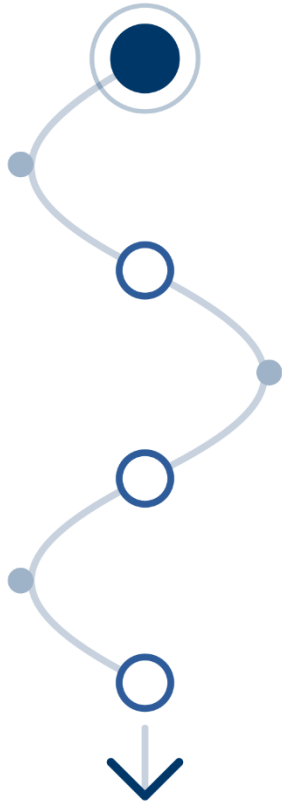
Working with Generative AI

From prompts to agents, and what it means for BA

Dawei CHEN

Week 4-2 | Thursday, 24 September 2026

Today



1. How you use AI, and where it fails

Poll, evidence, and a live demo

2. From prompts to loop engineering

Four layers of AI engineering, skills, and agents

3. AI in the BA workflow

Three interfaces; AI across the BA process

4. Fluent, not dependent

The 4D framework, overreliance, and the job market

Wrap-up: A1 due Monday, 28 September

Quick poll: what did you use AI for this week?

- ① Explaining a concept
- ② Writing or debugging code
- ③ Replying to emails
- ④ Application materials (CV, PS)
- ⑤ Something else

Show of hands

Raise your hand for every option that applies to you this week.

Then, open discussion

- The **most useful thing** AI did for you.
- One time it got something **wrong**.
How did you notice?

How others use AI

ChatGPT, individual accounts

[*OpenAI Signals, messages from June 2026*](#)

- 2×** more “doing” at work than outside work
- 45%** of work messages are “doing”: writing, coding, analysis
- 22%** of non-work messages are “doing”; asking leads

Claude, chat and Cowork

[*Anthropic Economic Index, April–June 2026*](#)

- 17%** of conversations produce an explanation
- 15%** produce a document or report
- 21%** of coursework produces a document; 20% an explanation
- ~50%** personal use at weekends, vs ~35% on weekdays

*AI is shifting from answering to doing. **Who is doing the thinking?***

Limitations of generative AI



Knowledge cutoff

Training data has a fixed end date; does not know recent events



Non-deterministic output

Same prompt can produce different results each time



Hallucination

Generates plausible-sounding but factually incorrect content



Context window

Limited amount of text the model can process at once



Reasoning limitations

Can struggle with multi-step logic, math, and causal reasoning

Source: [*Anthropic, AI Fluency: Framework & Foundations \(Capabilities & limitations; key terms\)*](#)

Live demo: how many of 30 postings ask for AI skills?

1 Ask the chatbot

___ / 30

Paste all 30 postings;
ask how many require
AI skills.

2 Naive code

___ / 30

`contains("ai")`
Search for the letters
“ai”.

3 Better keywords

___ / 30

AI, LLM, ChatGPT,
prompt, 大模型, 机器
学习 ...

4 Hand-coded

___ / 30

Apply the codebook to
each posting.

Define what you are counting before you count it.

A total close to the truth can still hide wrong labels: errors cancel out.

Data: 30 synthetic postings written for this class (job_postings_30.csv).

Codebook: what counts as an AI skill?

Code	Definition	Example
BUILD	Builds, trains, or evaluates ML or AI models, or LLM applications	“Fine-tune recommendation models; build RAG prototypes”
USE	Uses generative AI tools in the work itself	“Use ChatGPT to draft campaign insight summaries”
NONE	No AI skill required of the role	“Automate recurring reports with VBA”

- Required and preferred both count
- An AI company or industry does not count
- Automation and statistical models are not AI here

Which rule would you change?

Every change moves the count.

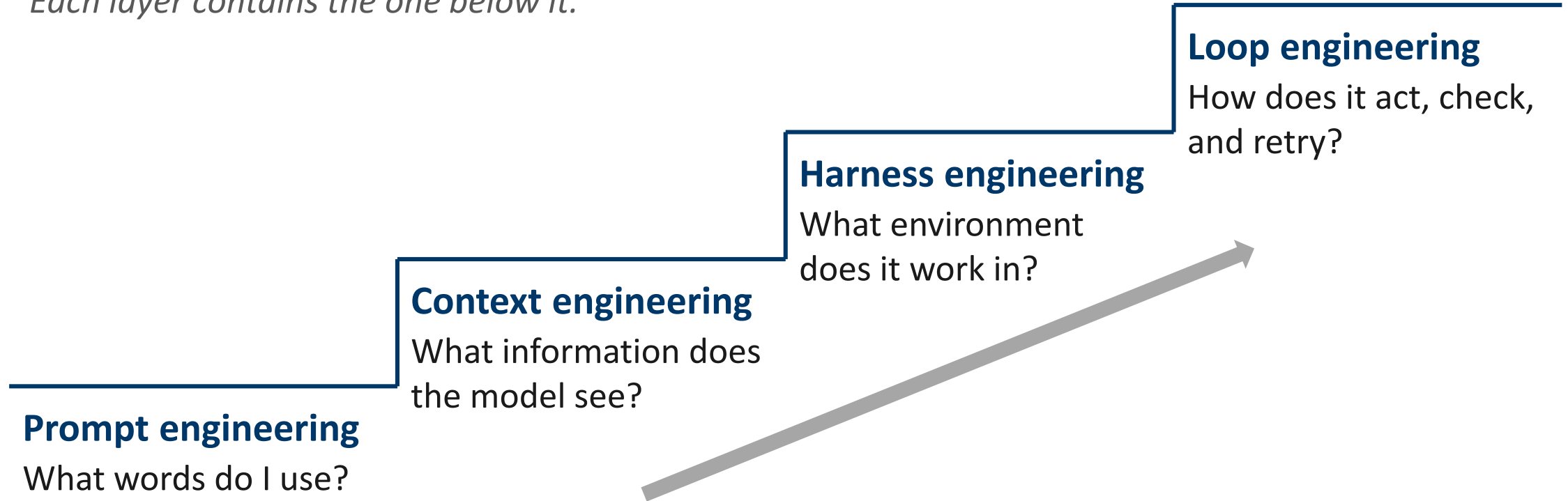
Personal use vs. engineering

	Personal use	Engineering
Unit of work	One chat	A repeatable system
Who checks	You, by eye	Built-in checks and tests
Who relies on it	Only you	Your team, manager, or client
When it fails	You try again	It can fail silently, at scale

For assignments in this course: closer to the left-hand or right-hand column?

Four layers of AI engineering

Each layer contains the one below it.



Sources: [Anthropic \(Sep 2025\) context engineering](#); [OpenAI \(Feb 2026\) harness engineering](#); [IBM \(Jul 2026\) loop engineering](#).

Layer 1: prompt engineering

*Ask the AI for help
with prompting*

Weak

Parse the salary
column.



Strong

Task: convert each salary string into minimum and maximum monthly pay (CNY) and months paid per year.

Format: CSV with columns raw, min, max, months.

Examples:

15-25K · 14薪 → 15000, 25000, 14

8千-1.2万 → 8000, 12000, 12

面议 → blank, blank, blank

Four elements: clear task · relevant context · examples · output format

Source: [Anthropic, Prompt engineering overview \(Claude documentation\).](#)

Layer 2: context engineering

“which words?” → “**what information does the model need to see?**”

Bare prompt

“Write a scraper for BOSS 直聘 job listings.”

Result: plausible code with invented selectors.

Engineered context

- Saved HTML of one real listing page
- Target schema: title, salary_min, salary_max, city, experience, education, skills
- Your coding scheme for skills
- Three hand-coded example records

Result: code that matches the real page.

Source: [Anthropic \(2025\), Effective context engineering for AI agents.](#)

Layer 3: harness engineering

“Humans steer. Agents execute.”

OpenAI engineering team, 2026

The engineer’s main job becomes designing the environment, specifying intent, and building feedback loops so agents can work reliably.

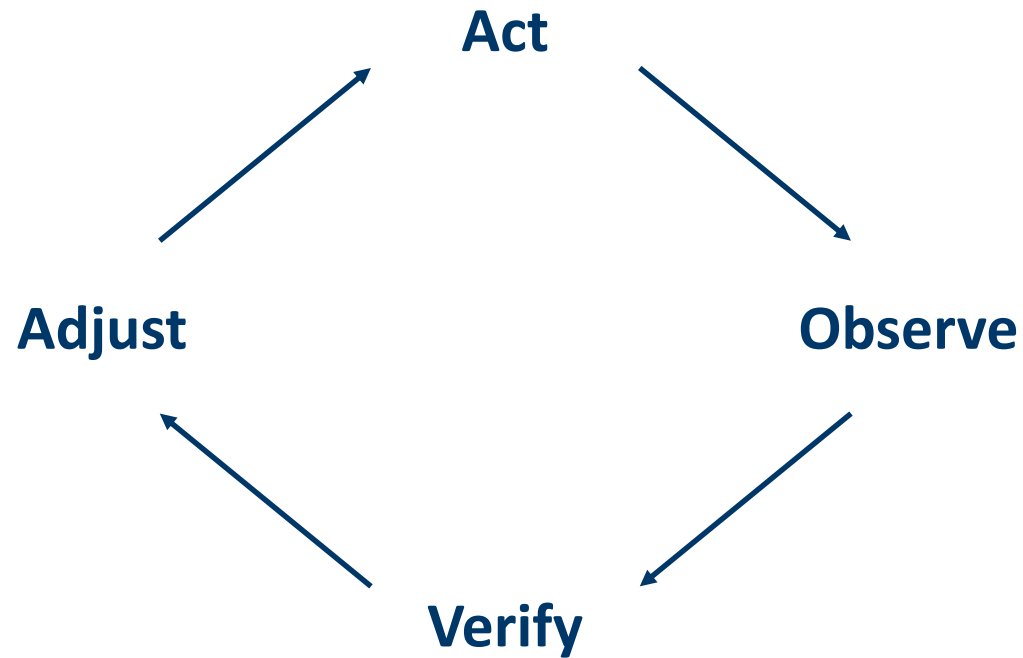
When the agent fails, ask: what capability or rule is missing from the environment?

```
al-project/  
├── README.md      intent, rules  
├── schema.md      field meanings  
├── checks.py      must pass  
├── scraper/  
└── data/raw/
```

Your repository is a harness: any AI-written code must pass checks.py before it is merged.

Source: [OpenAI \(2026\), Harness engineering: leveraging Codex in an agent-first world.](#)

Layer 4: loop engineering



Design four things

- **Goal:** what counts as done
- **Check:** how success is tested
- **Stopping rule:** when to give up
- **Budget:** a cap on time, tokens, and cost

Example: scrape → run checks.py → fix → repeat, until the checks pass or after five attempts.

Sources: [IBM \(2026\), What is loop engineering?](#); [Lulla et al. \(2026\)](#)

Skills: reusable know-how

Folders of instructions, scripts, and resources that the model loads only when a task needs them.

- Write the procedure once; reuse it every time
- Put the deterministic part in a script, not in prose
- Version it in your repository like code
- Use only skills you wrote or trust: a skill can run code

```
skills/salary-parsing/  
├── SKILL.md           when and how  
├── parse_salary.py    the rules  
└── examples.csv       20 cases
```



Source: [Anthropic, Agent Skills overview \(Claude documentation\)](#)

Workflows vs. agents

	Workflow	Agent
Next step decided by	Your code (a fixed path)	The model
Best for	Well-defined, repeatable tasks	Open-ended tasks
Trade-off	Predictable and cheaper	Flexible; costlier; errors compound
Project example	scrape → clean → extract → check	“Find why the salaries look wrong”

A single chatbot answer is neither. **Design rule: start with the simplest option that works.**

Sources: [Anthropic \(2024\), Building effective agents](#); [OpenAI, A practical guide to building agents](#).

Case: OpenAI's in-house data agent

What it is

- An internal agent that lets staff in engineering, finance, go-to-market, and research ask data questions in plain language

What was hard

- Finding the right tables among tens of thousands, not writing SQL
- Silent errors: many-to-many joins, filter mistakes, unhandled nulls

Source: [OpenAI \(2026\), Inside OpenAI's in-house data agent.](#)

Lesson

The value lies in context and verification, not in generating code.

Discussion (2 min)

What would an agent need to know to answer “Which cities pay BA roles most?” from your team’s data?

One model, three interfaces

	Chat	Coding agent	API
Who drives	You, message by message	The agent, within permissions	Your Python code
Where it runs	Browser or app	Your terminal and repo	Your script or pipeline
What it can touch	What you paste in	Files, commands, git	What your code sends
Use it for	Explaining, drafting, one-off help	Building and fixing code	Processing thousands of records the same way

Examples: Claude, Claude Code, Claude API (Anthropic documentation). The categories are vendor-neutral.

Same task, three ways

Task: classify 500 postings as DA, BA, DS, or other

Chat	Paste 20 postings and read the labels.	<i>Good for exploring the categories</i>
Coding agent	It writes and runs a classification script over the repo.	<i>Good for building the pipeline</i>
API	A fixed prompt, JSON output, every call logged.	<i>Reproducible and checkable</i>

We build the API version in Text analytics II (13 October).

Generative AI across the BA process

Stage	AI helps with	You must verify	Course link
Framing	Drafting candidate questions	Is it the real question?	Project launch
Acquisition	Writing scrapers	Row counts, pagination, legality	Web scraping
Cleaning	Parsing messy fields	Spot-checks against the source	A1, A2
Text → variables	Extraction, classification	A hand-coded gold set	Text analytics
Diagnosis	Suggesting methods, code	Assumptions, confounders	Causal inference
Prescription	Setting up the model	Constraints fit the business	Optimisation
Communication	Drafting slides and memos	Every number traces to a script	Final report

AI **adds least** where the context sits outside the data: framing, data judgement, causal assumptions.

Team exercise: where does your pipeline trust AI?

Open your A1 pipeline. Beside each step, mark:

- D** delegated to AI
- C** the check you use
- ?** no check yet

4 minutes

Before you leave: name one “?” that you will fix before A1 is due on Monday.

The 4D AI Fluency framework

AI fluency means using AI in ways that are:

Effective

Get the right outcome

Efficient

Use time & effort well

Ethical

Keep humans in the loop

Safe

Avoid harm & misuse

Competency	What it means	Where we met it today
Delegation	Deciding what AI should and should not do	AI across the BA process
Description	Communicating the task clearly	Prompt and context engineering
Discernment	Judging outputs for accuracy and quality	Limits and the live demo
Diligence	Owning the output; disclosing AI use	Your disclosure statement

Prompting is only part of one competency out of four.

Source: [Dakan, Feller & Anthropic \(2025\), AI Fluency: Framework & Foundations, Claude Academy](#)

Writing is thinking

vs.

Thinking is not only writing

The evidence on overreliance

17%

lower quiz scores for developers who learnt a new Python library with AI help

- Largest gap: debugging
- No significant time saving on average
- Those who asked conceptual questions kept learning

319

knowledge workers surveyed about AI and critical thinking

- More confidence in AI, less critical thinking
- More self-confidence, more critical thinking
- Thinking shifts towards verifying and integrating

Sources: [Shen & Tamkin \(2026\), How AI impacts skill formation, Anthropic](#); [Lee et al. \(2025\), CHI '25](#).

Working rules for this course

- 1 Ask for explanations, not only code.**
Conceptual questions protect learning.
- 2 Write the check before you accept the output.**
Counts, spot-checks, gold sets.
- 3 Disclose every use.**
Which tool, for what, and what you changed.
- 4 Practise without AI.**
It is not permitted in the midterm or mock interviews.



Generative AI and the entry-level job market

19%

Employment gap for 22–25-year-olds in highly AI-exposed US occupations, relative to less-exposed peers of the same age (June 2026)

- No economy-wide job displacement so far
- Fewer young hires, rather than more dismissals
- Declines where AI automates tasks; flat or rising where it complements workers
- Hit hardest: codified knowledge. Gaining: tacit knowledge from practice

Descriptive, not causal: the authors' own caution.

Does the same hold in China? Tracks 1 and 2 can test it.

Sources: [Brynjolfsson, Chandar & Chen \(Aug 2026\)](#); [Massenkoff & McCrory \(2026\)](#)

What “AI-native” means

AI-native PhD students

A label used widely in hiring, but rarely defined.

AI-native firm

Built around AI capabilities, through two channels:

- **Process:** AI changes how people work inside the firm
- **Product:** AI is built into what the firm sells

Such start-ups are 25% smaller, with ~15% fewer entry-level staff.

AI-native worker

“Reflexive AI usage is now a baseline expectation.”

Tobi Lütke, CEO, Shopify, April 2025

- AI use enters performance reviews and hiring
- Teams must show why AI cannot do a task before asking for staff

Sources: Fast Company (2025); [Kim & Koning \(2026\), AI-Native Firms, HBS WP 26-090](#); Lütke (2025).

The AI-native business analyst

Working definition (ours): directs AI across the workflow; owns framing, verification, and translation.

Capability	What it means for a BA	Evidence
Directs AI	Delegates to AI tools and agents; reviews what they produce	Microsoft Work Trend Index (2025) Every employee as an “agent boss”
Frames the problem	Turns a vague business question into an analysable one	WEF Future of Jobs Report (2025) Analytical thinking: top core skill
Verifies	Checks AI output before it informs a decision	Lee et al. (2025), CHI Effort shifts to verification
Translates	Turns analysis into decisions that managers act on	Henke, Levine & McNerney (2018) The “analytics translator” role
Brings context	Supplies the domain knowledge the data lacks	Anthropic Economic Index (2026) Users: AI lacks judgement, context

What this means for your applications

What to expect or show

Interviews

“How do you know this number is right?”; coding without AI

Portfolio

A repository with checks and an AI disclosure is evidence of judgement

Statement of purpose

Describe the decisions you made, not the tools you used

Positioning

AI reproduces codified skills; show framing, validation, and communication

These are hypotheses. Your project data will test them.

Q&A

Dawei CHEN | CBPM B214 | dchen@kean.edu

Office hours: Tue 16:00 – 17:15 | Wed 15:00 – 16:00 | Thu 14:15 – 17:00

Due: Monday, 28 September, 23:59 — A1: project brief and pilot data

Next session: Tuesday, 29 September — Consultation 1 – collection gate